Cement: history and production

Notes from the lecture slides “Module 1a – Cement properties and testing (Cement – production, composition and testing)”

In this module

1. Binding materials before Portland cement
2. Portland cement as an industrial product
3. Raw materials
4. Manufacturing process
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1 Binding materials before Portland cement

Long before modern cement, builders used three kinds of binders:

- **Lime-based materials**, made from limestone and gypsum. These were the main binders.
- **Pozzolans**, volcanic ash and similar materials, often used without the builders knowing why they worked.
- **Bitumen**, a natural tar.

The word *concrete* comes from the Latin *concretus*, meaning "to grow" or "grown together". A **hydraulic cement** is one that sets and hardens even under water, because it hardens by reacting with water rather than by drying.

1.1 Smeaton and the Eddystone lighthouse (1756)

- While planning the **Eddystone lighthouse** tower in 1756, **John Smeaton** found that the best limes for mortar were made from limestone containing a

high proportion of clay.

- This was the key discovery: **cement = limestone + clay**, burnt together.
- He finally used this hydraulic lime mixed with **pozzolana in equal quantities**.
- The tower was completed in **1759**, was 72 feet tall with 93 steps, and lasted until a new lighthouse was built in 1882. Smeaton's tower was then moved stone by stone to **Plymouth**, where it still stands.

1.2 From hydraulic lime to Portland cement

- **L. J. Vicat** made an *artificial hydraulic lime* by calcining (burning) an intimate mixture of limestone (chalk) and clay. This was the main forerunner of Portland cement.
- In **1824, Joseph Aspdin** patented his hydraulic cement and named it **Portland cement**, because the hardened product looked like **Portland stone**, a high-quality, durable limestone from Dorset, UK. This was *"proto" Portland cement*.
- His son **William Aspdin** later produced *"meso" Portland cement*, burnt at a higher temperature. Microscope images of his clinker show **alite (C_3S)** and **belite (C_2S)** crystals.
- Development therefore went **Proto → Meso → Normal (modern) Portland cement**.

Parameter	Aspdin clinker	Modern clinker
Relative burning rate	Slow (poor)	Quick
Alite (C_3S) crystal size	~60 µm	10–40 µm
Belite (C_2S) crystal size	5–10 µm	20–60 µm
Cooling rate	Slow (poor)	Quick
αC_2S content	Nil	10–20%
Kiln	Vertical (beehive)	Rotary

Cement chemists' shorthand

Cement chemistry uses one-letter symbols for the oxides: **C** = CaO, **S** = SiO_2, **A** = Al_2O_3, **F** = Fe_2O_3 (and later **H** = H_2O, **$\bar{S}$** = SO_3). So C_3S means $3CaO{\cdot}SiO_2$ (tricalcium silicate, *alite*) and C_2S means $2CaO{\cdot}SiO_2$ (dicalcium silicate, *belite*).

2 Portland cement as an industrial product

Portland cement is an unusual industrial product. It is made in huge quantities in special plants that **can make nothing else**, through a chain of unusual unit operations:

1. mining and very fine-scale blending of raw materials
2. very high temperature clinkering reactions
3. controlled cooling
4. grinding
5. blending
6. shipping under controlled conditions

Despite the huge tonnages, the **chemical composition is held within narrow limits**.

2.1 Economics of a cement plant

- A plant costs upwards of **$250 million**, a large fixed investment.
- It must run **continuously** (three shifts a day) both to pay off this capital cost and to **maintain kiln integrity**; heating and cooling the kiln lining repeatedly damages it.
- It must meet **strict environmental limits** (dust, CO_2, fuel emissions).
- And all this is done to make a commodity that sells for only about **Rs. 6 per kg**.

2.2 World production

- In 2018, **China** made about **53%** of the world's cement and **India** about **8%** (second largest). The rest of the world together made about 24%, with countries such as the EU, Vietnam, USA, Indonesia and Turkey at 1–3% each.

- World production rose steadily from about **1.5 billion tonnes in 1998 to about 4.2 billion tonnes in 2014**.
- Cement production has grown alongside fossil-energy production through the 20th century, a reminder that cement making is very energy-intensive.

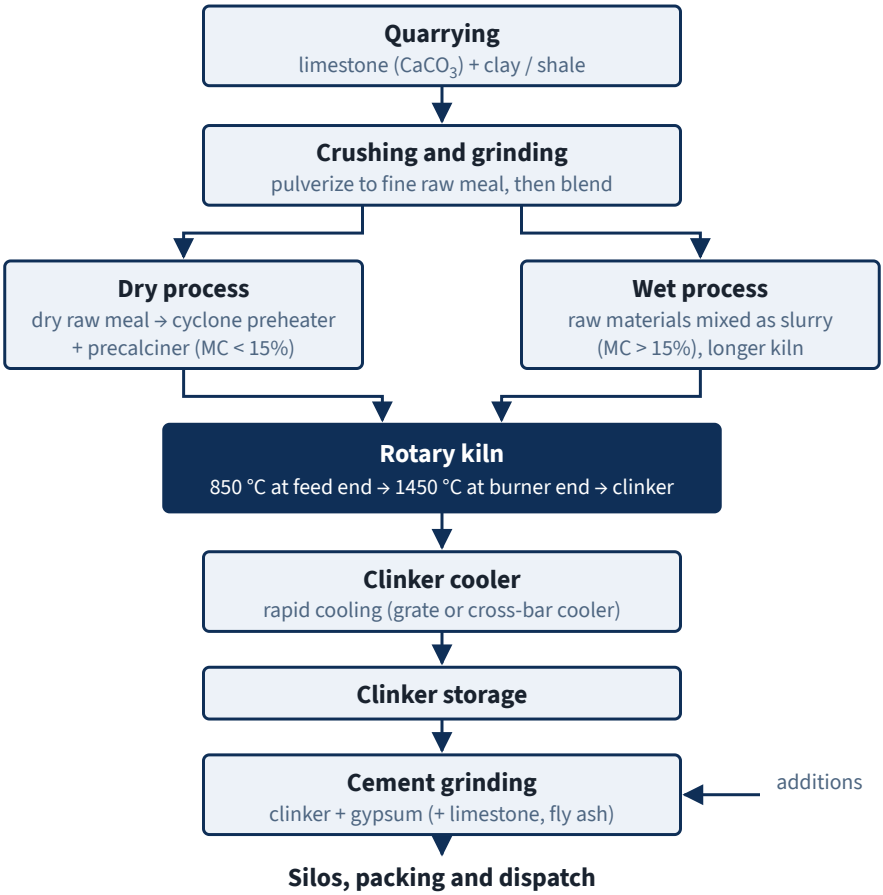
3 Raw materials

Type	What it supplies	Examples
Calcareous material	$CaCO_3$, the source of lime (CaO)	Limestone (main source); may contain some iron and alumina as impurities
Argillaceous material	Clayey matter: SiO_2 and Al_2O_3	Clay, shale
Gypsum	$CaSO_4 \cdot 2H_2O$	Added only at the final grinding stage, as a **set regulator**
Mineral additions	Filler or supplementary material	Ground limestone or fly ash; IS permits up to **5%** in OPC

The four oxides needed can come from many sources:

Calcium	Silicon	Aluminium	Iron
Limestone	Clay	Clay	Clay
Marl	Marl	Shale	Iron ore
Calcite	Sand	Fly ash	Mill scale
Aragonite	Shale	Aluminium ore refuse	Shale
Shale	Fly ash		Blast furnace dust
Sea shells	Rice husk ash		
Cement kiln dust	Slag		

4 Manufacturing process



Steps in Portland cement manufacture, from quarry to dispatch.

4.1 Pulverization

The raw materials are crushed and ground to the right fineness before burning, because:

- **finer material blends better** and **burns more completely**, and
- it **reduces the overall power (fuel) consumption** of the kiln.

4.2 Blending: wet or dry process

The ground raw materials are blended either as a dry powder (**dry process**) or as a slurry with water (**wet process**).

	Wet process	Dry process
Mixing	More uniform mixing (slurry)	Less uniform, but adequate with modern controls
Kiln length	Long, about **270 m**	Short, about **50 m**
Output	Lower	**Higher**
Power and fuel	Higher (the water has to be evaporated)	**Lower**
Use today	Being phased out	**Dry process with precalciners is the standard today**

The choice depends mainly on the **moisture content (MC)** of the raw materials:

- **MC > 15%**: wet blending (slurry) is preferred.
- **8% < MC < 15%**: dry blending **with precalciners**.
- **MC < 8%**: dry blending.

4.3 Preheaters and precalciners

- A **preheater** is a vertical stack of **cyclone separators** (typically four stages). The raw meal falls down through the cyclones while hot kiln exhaust gases rise up through them, so the meal is **heated uniformly** and heat from the exhaust is recovered.
- In a **precalciner**, fuel is also burnt inside the cyclone system. This lets the **calcination reaction ($CaCO_3 \rightarrow CaO + CO_2$) partly complete before the material even enters the kiln**, so the kiln can be shorter and output higher.

4.4 Burning in the rotary kiln

- Only **rotary kilns** are used today: long steel tubes lined with refractory, slightly inclined and slowly rotating.
- Typical size: **30–70 m long and 5–6 m in diameter**. The length also depends on the blending process; wet-process kilns are much longer.

- The temperature rises from about **850 °C at the inlet (feed end) to 1450 °C at the outlet (burner end)**.
- Not all reactions finish inside the kiln; **some happen during cooling**.

> **Remember**
>
> The product that comes out of the kiln, as dark grey nodules, is called **clinker**.

5 Reactions in the kiln

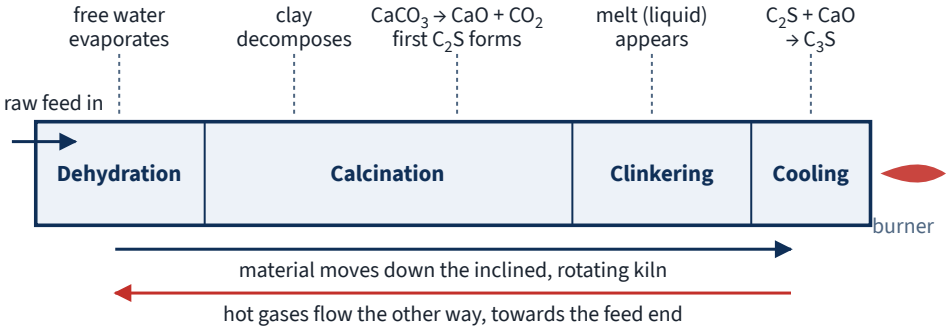
5.1 What clinkering does

The clinkering reactions convert a mixture of calcium carbonate and silica- and alumina-bearing materials into **special crystalline compounds that can react with water** to give controlled setting and strength gain.

- **Major components:** impure, well-crystallized, fine (under about 50 µm) crystals of **tricalcium silicate (C_3S, alite)** and **dicalcium silicate (C_2S, belite)**.
- **Minor but important components:** extremely fine crystals of **tricalcium aluminate (C_3A)** and **calcium aluminoferrite solid solution (C_4AF, ferrite)**.
- **Present in small amounts but very important:** deposits of soluble crystalline compounds on the clinker surfaces, namely **alkali sulfates and calcium alkali sulfates**. Because they dissolve quickly, they strongly influence early setting.

5.2 Reactions at each temperature

Temperature	What happens
Up to 700 °C	**Activation of silicates**: free and combined water driven off; changes in the crystal structure of clay minerals
700–900 °C	**Decarbonation of $CaCO_3$** ($CaCO_3 \rightarrow CaO + CO_2$); first combination of alumina, iron oxide and activated silica with lime
900–1200 °C	**Belite (C_2S) forms**
Above 1250 °C (mainly above 1300 °C)	A **liquid (melt) phase appears**. It speeds up the reaction of belite with free lime to form **alite (C_3S)**: $C_2S + CaO \rightarrow C_3S$
Cooling	The melt, which contains the C_3A and C_4AF, **turns to glass** if cooled quickly



Feed	50 °C	600 °C	1000 °C	1350 °C	1450 °C
Gas	450 °C	800 °C	1200 °C	1350 °C	1550 °C

Typical temperatures along the kiln (after Mindess and Young, 1981)

Zones of a rotary kiln. The raw feed enters at the cooler, upper end; the flame is at the lower end.

The kiln is often described in **four zones**:

1. **Dehydration zone**: water is removed (feed up to about 600 °C).
2. **Calcination zone**: clay and then limestone decompose; CO_2 is released; initial compounds and C_2S begin to form (feed about 600–1000 °C).
3. **Clinkering (burning) zone**: melt forms and C_3S forms (feed about 1350–1450 °C; gas up to about 1550 °C).

4. **Cooling zone**: near the outlet, before the clinker drops into the cooler.

> **Why the cooling rate matters**
>
> If clinker is cooled **slowly**:
>
> - **C_3A crystallizes out** of the melt instead of staying as glass, which causes **setting problems** (rapid, uncontrolled setting); and
> - **alite can decompose back to belite and free lime** ($C_3S \rightarrow C_2S + CaO$), lowering early strength and leaving free lime that can cause unsoundness.
>
> So modern plants cool clinker **quickly**.

6 Clinker coolers

Hot clinker leaving the kiln is cooled rapidly in a cooler. Two common types:

- **Grate cooler**: clinker travels along a moving grate while cold air is blown up through it.
- **Cross-bar cooler**: a newer design in which cross-bars move the clinker bed over a fixed air-distribution floor.

The hot air from the cooler is reused as combustion air in the kiln, saving fuel.

7 Intergrinding with gypsum

This is the **final step** in cement manufacture.

- Clinker is ground together with about 3–5% **gypsum**, which acts as a **set regulator**. Without gypsum, the C_3A reacts almost instantly with water and the cement **flash sets** (stiffens immediately with heat release).
- The **mill temperature must be strictly controlled**. If the mill gets too hot, gypsum loses water and turns into hemihydrate, which can cause *false set* in the cement.
- Grinding was traditionally done in **ball mills**; **vertical roller mills/presses** are now used for better energy efficiency.

- The cement is ground to the **required fineness**, which controls the rate of hydration and strength gain.

8 Quality control

Cement manufacture today is highly controlled, yet quality still varies **between brands, within the same brand, and even between batches made on the same day**. So quality is checked at **every stage**:

1. after excavation from the quarry,
2. before and after blending the raw feed,
3. after clinker is formed,
4. after intergrinding clinker with gypsum, and
5. finally, before packing into bags or bulk containers.

8.1 Quality control parameters

Here C, S, A and F are the percentages of CaO, SiO_2, Al_2O_3 and Fe_2O_3 in the raw mix or clinker.

Lime saturation factor (LSF)

$$\text{LSF} = \frac{\text{C}}{2.8\,\text{S} + 1.2\,\text{A} + 0.65\,\text{F}}$$

- Usually **92–98%**.
- LSF **above 100%** means more lime than can combine with the other oxides, so **free lime** will remain in the clinker.

Silica ratio (silica modulus)

$$\text{SR} = \frac{\text{S}}{\text{A} + \text{F}}$$

Usually **2.0–3.0**. A higher SR means less melt in the kiln, so the mix is harder to burn.

Alumina ratio (alumina modulus)

$$AR = \frac{A}{F}$$

Usually **1.0–4.0**. It controls the proportions of C_3A and C_4AF in the clinker.

- The **potential C_3S content** is also checked, using the Bogue equations (covered in Module 1b).
- **LSF is the most important** parameter because it decides **how much free lime** will be present. **Too much free lime makes the cement unsound**: free lime hydrates slowly after the concrete has hardened, expands, and cracks it.
- Indian Standards write the LSF with a sulfate correction in the numerator, (CaO − 0.7 SO_3), but the idea is the same.

Worked example

A raw mix has CaO = 65%, SiO_2 = 21%, Al_2O_3 = 5.5% and Fe_2O_3 = 3.5%. Find LSF, SR and AR.

- LSF = 65 / (2.8 × 21 + 1.2 × 5.5 + 0.65 × 3.5) = 65 / (58.8 + 6.6 + 2.275) = 65 / 67.675 = **0.96, i.e. 96%**, within 92–98%, so little free lime is expected.
- SR = 21 / (5.5 + 3.5) = **2.33**, within 2.0–3.0.
- AR = 5.5 / 3.5 = **1.57**, within 1.0–4.0.

Quick revision

- Hydraulic cement hardens by reacting with water, even under water. Smeaton (1756) showed lime from clayey limestone is hydraulic; Aspdin named Portland cement in 1824.
- Raw materials: calcareous (limestone, $CaCO_3$) + argillaceous (clay: SiO_2, Al_2O_3); gypsum added at grinding as set regulator; IS allows up to 5% mineral additions in OPC.

- Process: quarry → crush and grind → blend (wet if MC > 15%, dry if < 8%, dry with precalciner between) → preheater/precalciner → rotary kiln (850 → 1450 °C) → clinker → rapid cooling → grind with gypsum → dispatch.
- Kiln: dehydration → calcination ($CaCO_3 \rightarrow CaO + CO_2$, 700–900 °C) → C_2S forms (900–1200 °C) → melt and C_3S form (> 1250–1300 °C) → cooling.
- Slow cooling: C_3A crystallizes (setting problems) and C_3S breaks down to C_2S + free lime.
- No gypsum → flash set. Hot mill → gypsum dehydrates → false set.
- LSF = C/(2.8S + 1.2A + 0.65F), 92–98%; SR = S/(A+F), 2–3; AR = A/F, 1–4. LSF > 100% → free lime → unsoundness.

Questions to practise

1. What is a hydraulic cement? Briefly trace the history of Portland cement from Smeaton to Aspdin. (5 marks)
2. Name the raw materials used to make Portland cement and the oxide each one supplies. (3 marks)
3. Compare the wet and dry processes of cement manufacture. How is the process chosen? (5 marks)
4. With a neat flow chart, describe the manufacture of Portland cement by the dry process. (10 marks)
5. What are preheaters and precalciners? Why are they used? (3 marks)
6. Describe the reactions taking place in the rotary kiln at different temperatures, with a sketch of the kiln zones. (10 marks)
7. Why must clinker be cooled rapidly? (2 marks)
8. Why is gypsum added to cement clinker? What happens if the grinding temperature is too high? (3 marks)
9. Define lime saturation factor, silica ratio and alumina ratio, and give their usual ranges. Why is LSF the most important? (5 marks)

10. A raw mix contains CaO 64%, SiO_2 22%, Al_2O_3 5% and Fe_2O_3 3%. Calculate the LSF, silica ratio and alumina ratio and comment on the result. (5 marks)